

Data Schema (ERD)

AI-Powered Content Operations Platform — Skip Studio

Prepared for	Skip Studio (fictional client — Exology Pioneer Program, Week 4 Final Project)
Prepared by	Amr Hassan Safieddine Nagy
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Status	Draft — for review

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What this describes

The things the product stores: clients and the market they operate in, the people who work on them, the brand and compliance rules that govern their content, the regional calendar their content plans around, the campaigns and every multi-form deliverable they produce, the two-stage approval history behind each one, the Instagram account each client is connected through, the performance data that comes back once something is live, and the requests and comments a client raises directly. Built for full traceability: every scheduled item can be traced back to the brief it came from, the client and market it belongs to, the brand guide version it was grounded in, the rule it cited, and the exact approval history that cleared it to publish.

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Entities

People & access

► User

user_id	PK
name	
user_type	staff client_contact
is_agency_admin	staff only; assigns roles on client records

► ClientAssignment — per-client team, beyond the default account manager

assignment_id	PK
client_id	FK → Client, user_id
FK → User	
role_on_client	content_lead (replaces account manager as reviewer) content_creator client_approver

A client_approver User may have exactly one ClientAssignment row, ever — enforced by a unique constraint on user_id. Staff roles are intentionally many-to-many with Client; a client contact is not.

Clients & markets

► Client — the client roster

client_id	PK
name , industry	
status	active inactive — gates whether any content may be produced
tier	informational only, no behavior tied to it
channels	list
market_id	FK → Market
single-select, not nullable once set	
account_manager_id	FK → User
default internal reviewer for this client	
active_brand_guide_id	FK → BrandGuideVersion, nullable
null = no governing guide on file	
sensitive_sector	derived from industry (healthcare / financial / government)

► **Market** — seeded with exactly one row today

market_id	PK
name	"Saudi Arabia"
country_code	SA
calendar_system	informational — e.g. "gregorian_and_hijri"

A real table rather than a hardcoded enum, so a second market is a data insert, not a migration.

Brand governance

► **BrandGuideVersion**

brand_guide_version_id	PK
client_id	FK → Client
version_number	
status	draft pending_client_approval active superseded
created_by_id	FK, client_approved_by_id FK, nullable
created_at	, approved_at

Only one version per client is ever active. Past versions stay superseded, never deleted.

► **GuidelineClause** — the agency standards and every client's brand rules

clause_id	PK
source_type	agency brand
brand_guide_version_id	FK, nullable null for agency clauses — global, unversioned
clause_code	, title, text e.g. 0.6, CR.4, NB.4

Calendar

► **Occasion** — regional and local calendar entries, scoped to one market

occasion_id	PK
market_id	FK → Market
name	, category e.g. "Ramadan"; religious national seasonal retail
date_type	fixed_gregorian hijri_based
month	, day used only when date_type = fixed_gregorian

► **OccasionDate** — resolved per-year dates for occasions whose Gregorian date moves

occasion_date_id	PK
occasion_id	FK → Occasion only populated for hijri_based occasions
year	, gregorian_date the resolved date for that year
source	seeded — hand-maintained, not calculated

Campaigns & content

► **Campaign** — *the brief, and the campaign it becomes*

campaign_id	PK	
client_id	FK → Client	
title, objective, audience, channels, raw_brief_text		as submitted
related_occasion_id	FK → Occasion, nullable	set when built around a specific upcoming occasion
submitted_by_id	FK → User	
status		received info_requested in_progress complete
override_attempt_detected		true if the brief tried to skip or fake approval
compliance_review_required		true for sensitive-sector clients

► **ContentItem** — *one row per drafted deliverable*

content_item_id	PK	
campaign_id	FK → Campaign	
content_form		post image video reel photoshoot email blog_post ad_copy hashtag_set cta creative_prompt
platform		instagram tiktok facebook linkedin email
content_body		current draft text; visual/video forms carry content via linked MediaAsset rows instead
scheduled_date		target/proposed publish date — distinct from the scheduled status
status		drafted in_refinement pending_internal_review internal_approved pending_client_review client_approved scheduled declined flagged publishing published publish_failed
flagged_clause_id	FK, nullable	primary rule broken, if flagged
parent_content_item_id	FK, nullable	self-reference for A/B variants
grounded_brand_guide_version_id	FK	version active at draft time — frozen even if the guide is edited later
created_at, updated_at		

► **ContentItemCitation**

citation_id	PK	
content_item_id	FK, clause_id	FK

Every drafted item can be grounded in more than one clause — an agency clause and a brand rule together.

► **MediaAsset** — *generated or uploaded images and video*

media_asset_id	PK	
content_item_id	FK → ContentItem	
asset_type		image video
generation_source		ai_generated uploaded — a photoshoot's output is always uploaded
storage_url, format		
created_by_id	FK → User	

► **ReferenceAttachment** — *reference material a creator supplies when prompting generation or regeneration of one item*

attachment_id	PK	
content_item_id	FK → ContentItem	the specific deliverable being generated or regenerated
uploaded_by_id	FK → User	must resolve to role_on_client = content_creator
file_type		image pdf doc
storage_url		
instruction		optional text accompanying the file, e.g. "match this angle"
created_at		

Image references are passed as vision context; PDF/doc references are text-extracted as context. Rows accumulate across regenerations rather than being overwritten. Carries no authority over compliance, same as brief wording and comments.

Review & compliance

► **Approval** — *one record per decision, per item, per stage*

approval_id	PK	
content_item_id	FK → ContentItem	
stage		internal client
decision		approve decline — an approval that requests a change is recorded as decline
comment		required if decline, optional if approve
decided_by_id	FK → User, decided_at	
bulk_action_id	, nullable	groups rows created by one "approve whole plan" click

The gate reads the most recent row per (content_item_id, stage) — not whether an approval exists anywhere in history. This is what lets a reviewer or client decline something they already approved. Decline is available at pending_internal_review/pending_client_review, and remains available at internal_approved, client_approved, and scheduled, for either the internal reviewer/content lead or the client. Decline stops applying once publishing or published.

► **Flag** — *anything routed to a human instead of drafted or scheduled*

flag_id	PK	
campaign_id	FK, content_item_id	FK, nullable
		null content_item_id = brief-level flag
clause_id	FK	the rule broken
flag_type		brand_violation compliance_violation unknown_client inactive_client cross_client_data approval_override_attempt
resolved	, resolution_notes	, resolved_at

► **AuditLog** — *append-only*

audit_id	PK	
entity_type	, entity_id	, action
		created edited scheduled rescheduled deleted approved declined flag_raised flag_resolved published take_down
performed_by_id	FK → User, performed_at	

Publishing & performance

► **PlatformConnection** — a client's connected Instagram account

platform_connection_id	PK
client_id	FK → Client
platform	instagram (enum kept open for future platforms)
access_token	encrypted at rest
status	connected expired disconnected
connected_by_id	FK → User staff only

No privacy field — Instagram Professional accounts cannot be private, so there is nothing to model.

► **MetricSnapshot** — time-series, not a single mutable row

metric_snapshot_id	PK
content_item_id	FK → ContentItem
captured_at, metric_type, value	impressions reach likes comments shares saves

Client requests

► **PostRequest** — the client-initiated calendar ask; carries no authority

post_request_id	PK
client_id	FK, requested_by_id
requested_by_id	FK → User (client_contact)
requested_date	the calendar day being asked for
related_content_item_id	FK, nullable set for "change/reschedule this existing post"
status	new under_review converted declined
linked_campaign_id	FK, nullable set once converted into a real brief

► **Comment** — discussion thread, distinct from Approval.comment

comment_id	PK
post_request_id	FK, nullable exactly one of these two is set
content_item_id	FK, nullable
author_id	FK → User, body

A Comment never triggers an approval reset — only a formal Approval decline or a deliberate edit does.

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PRD roles → schema representation

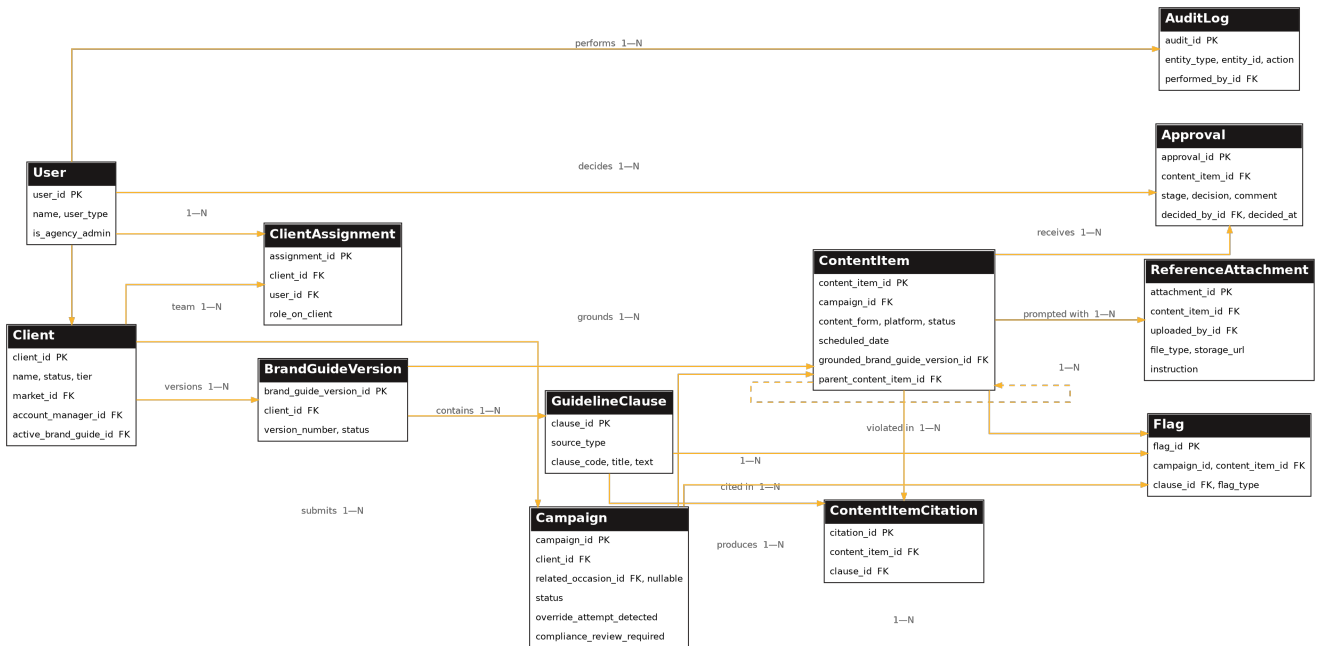
Five user types, none with a dedicated table — each is a User plus, for client-scoped roles, a ClientAssignment row. This is what lets Admin assign or reassign anyone by editing a row, with no separate user-management screen.

ROLE	USER_TYPE	ROLE_ON_CLIENT	NOTES
Account Manager	staff	— (Client.account_manager_id)	Submits briefs; is the internal reviewer unless a content lead is assigned; creates clients; manages PlatformConnection; converts PostRequests.
Content Creator (incl. Marketing Specialist)	staff	content_creator	Zero or more per client; authors the full multi-form plan; refines drafts pre-internal-review; attaches image/PDF/doc references when prompting generation or regeneration.
Content Lead	staff	content_lead	Optional; replaces the account manager as internal reviewer for that client, with the same late-revoke power.

Client	<code>client_contact</code>	<code>client_approver</code>	One or more per client. Approves/declines via the same Approval table, stage = client. Sees their assigned account manager; raises PostRequests and Comments; views their own analytics.
Agency Admin	<code>staff</code> , <code>admin=true</code>	—	Not tied to any one client; edits <code>account_manager_id</code> on Client and rows in ClientAssignment directly.

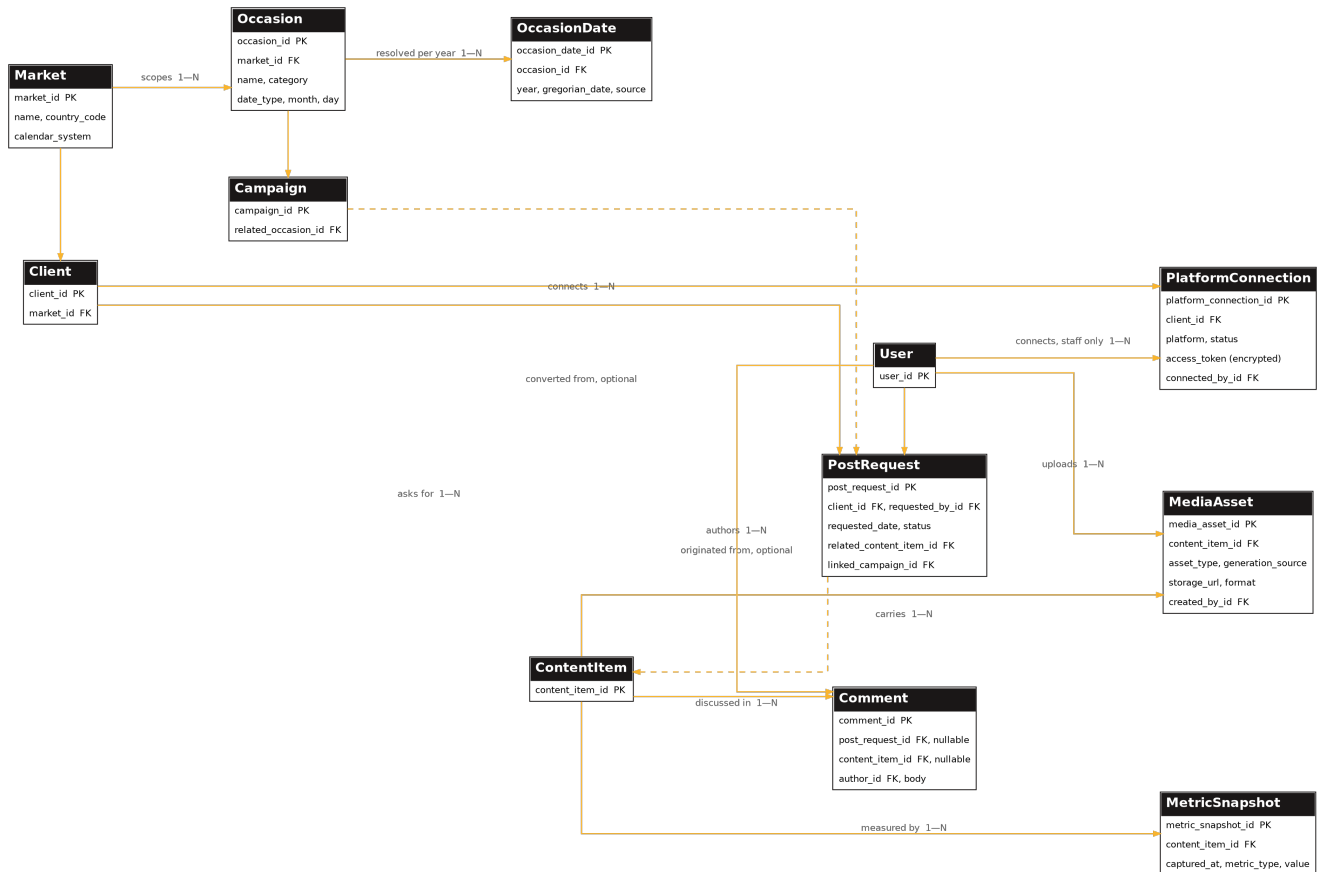
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Relationship diagram — client, campaign & content



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Relationship diagram — market, publishing & client requests



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Traceability

REQUIREMENT	SATISFIED BY
Choose market (Saudi Arabia for now)	Market table, Client.market_id, seeded with one row
Regional and local occasion calendar	Occasion + OccasionDate, scoped by market_id, Hijri-aware
Full multi-form content plan	ContentItem.content_form + MediaAsset
Schedules refer to upcoming events and occasions	resolve_calendar engine step + Campaign.related_occasion_id + scheduled_date
Publish to Instagram	PlatformConnection + Publishing Layer, scoped to one tester account
Performance analytics for Account Manager and Client	MetricSnapshot, time-series, role-scoped queries
Account Manager creates clients with market	Client create flow, market_id required
Client sees assigned account manager	Direct read of Client.account_manager_id
Client calendar request/schedule with comments	PostRequest + Comment, no authority over the gate
Edited approved or scheduled post resets approval	ContentItem.status reset on content or scheduled_date change
Client and reviewer can decline after approval or scheduling	Approval, gate reads most-recent-per-stage, symmetric for both actors
Nothing scheduled or published without both current approvals	Atomic gate re-check in the scheduler — dedicated race-condition test
A published post cannot be silently declined	status = published excluded from decline; separate take-down action
No client's content or guide visible to another client	Every content-bearing table scoped by client_id via campaign_id → Client
Creator can attach reference files for content generation	ReferenceAttachment, scoped to one content_item_id, creator-only, image/PDF/doc only